

The Editor-in-Chief
Journal of Modern Power Systems and Clean Energy

Dear Editor-in-Chief,

We submit the manuscript “*Baseline Configuration and Random-Sampling Controls in Metaheuristic Comparisons for Wind Farm Layout Optimization*” by Prince Solanki, Pulkit Dwivedi, Vanita Garg and Vinay Shukla for consideration as a regular paper in the *Journal of Modern Power Systems and Clean Energy*.

What the paper contributes. Many metaheuristics have been proposed for wind farm layout optimization, but their comparisons are hard to trust because baselines, budgets, feasibility and statistics are handled differently. The main contribution is methodological: a controlled evaluation protocol (equal budgets that count every objective and gradient evaluation, seed-paired runs, feasibility-aware ranking, Holm-corrected tests), equivalence analysis at three levels (seeds, cases and farm clusters) with the minimal equivalence margins reported, equal-cost random-sampling controls for two-phase hybrids, robustness to the direction resolution of the wind rose, supporting theory, diagnostics from instrumented reruns, and more than 200 automated checks that tie every number and claim of the text to the published data. We apply it to eight methods on the discretized Kusiak–Song benchmark (68 cases, 15° rose, 6,030 evaluations from random starts), a Horns Rev 1 block and IEA37 Case Study 1. Three findings follow. (M1) The PSO setting $w = 0.7$, $c_1 = c_2 = 2$ lies beyond the order-2 (variance) convergence boundary; in a sweep of $c_1 = c_2$, feasibility starts to degrade there ($c^* = 1.7486$) and falls progressively beyond it. With it, PSO ranks behind both salp-swarm hybrids (position 5) in the eight-method pool of our earlier comparison; with the constriction setting, it ranks first there. (M2) A salp-swarm (SSA) first phase adds nothing beyond equal-cost random sampling in the farm (a gain above 0.05 percentage points (pp) of wake loss is ruled out across seeds and cases), the Laplacian SSA as published is worse, and a PSO phase gives a clear gain; at this budget the second phase, variable neighborhood search (VNS), acts essentially as one compass-search descent. (M3) PSO followed by VNS (PSO-VNS) has the best average rank with 15° , 5° and 1° direction bins, also under a Gaussian wake at 1° , but its average advantage over a well-configured PSO is small: within ± 0.05 pp across seeds and cases, but not farm clusters, under the 15° rose, and a small but significant -0.043 pp (PSO-VNS minus PSO) under a 1° rose; case-level differences go both ways (11 cases favor PSO-VNS and 8 favor PSO by more than the margin). PSO-VNS reaches feasibility more reliably (Horns Rev 1 block: 30/30 vs. 19/30 runs, McNemar $p = 9.8 \times 10^{-4}$). It is a consistent reference method, not a superior optimizer.

Corrections made during the revision. We found an error in the wind-direction binning of our Horns Rev 1 model (the sectors received alternately seven and five bins), corrected it and repeated all Horns Rev 1 runs. We recomputed the PyWake reference at identical bins: our wake loss differs from that of PyWake’s NOJ model by -0.72 pp because our model takes the thrust coefficient at the free-stream speed; with the local thrust coefficient it reproduces PyWake exactly. We state plainly that with 1° bins none of the 901 feasible optimized Horns Rev 1 layouts exceeds the installed block; at 6,030 evaluations the Horns Rev 1 result is a feasibility result.

Scope and limitations. The results hold for discretized wind roses, a top-hat Jensen wake, circular farms, a $4D$ spacing and a prescribed number of turbines, with random starts and 6,030 evaluations unless stated. From feasible starts, VNS alone ranks first on the six largest cases; on IEA37 the best PSO-VNS layouts (30,030 evaluations) have AEP gaps of -1.93% (16 turbines) and -6.23% (36 turbines) to the best feasible published layouts; the pool lacks CMA-ES and L-SHADE; loads, costs and electrical losses are outside the scope, and all energy values are gross.

Fit to the journal. The layout fixes the energy yield of a wind farm for its lifetime, and layout optimizers are used at the planning stage to compare sites, turbine counts and spacing rules. Such decisions can rely on an optimizer only if its reported advantage survives a fair comparison. The paper gives planners and method developers a reproducible way to evaluate layout methods, a simple reference method, and checks of the conclusions against the power-curve, wake-model, direction-resolution and spacing assumptions. It continues the journal’s line of work on metaheuristic layout optimization.

Disclosure of an earlier submission. A related manuscript by the same authors, “*A Hybrid Laplacian Salp Swarm and Variable Neighborhood Search Algorithm for Continuous Wind Farm Layout Optimization*” (Manuscript ID [TBD: ID of the earlier MPCE submission]; decision: [TBD: e.g., rejected]), was previously submitted to the journal. The present manuscript is a new paper with a different thesis, and the 22 comments of the three reviewers shaped it directly. The requests for equal-budget comparisons with stronger baselines and proper statistics led us to recheck the PSO baseline, which revealed M1. The request for an ablation of the Laplace mechanism led to the component analysis with random-sampling controls, which revealed M2 and made us drop the salp-swarm phase. The requests for a real site and for tests of the power curve, wake model and 4D spacing led to the Horns Rev 1 and IEA37 cases and to the sensitivity analysis. A point-by-point mapping of all 22 comments is attached; it also names the points that are only partly addressed.

The manuscript is original, not published and not under consideration elsewhere; all authors approved its submission and declare no competing interests. Code and data: [TBD: Zenodo DOI].

Sincerely,

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